

Weather-dependent
renewable generation
available capacity

Connection point of
offshore wind farms
and hydrogen platforms

Nodal energy demands
(including power,
natural gas, hydrogen,
and ammonia)

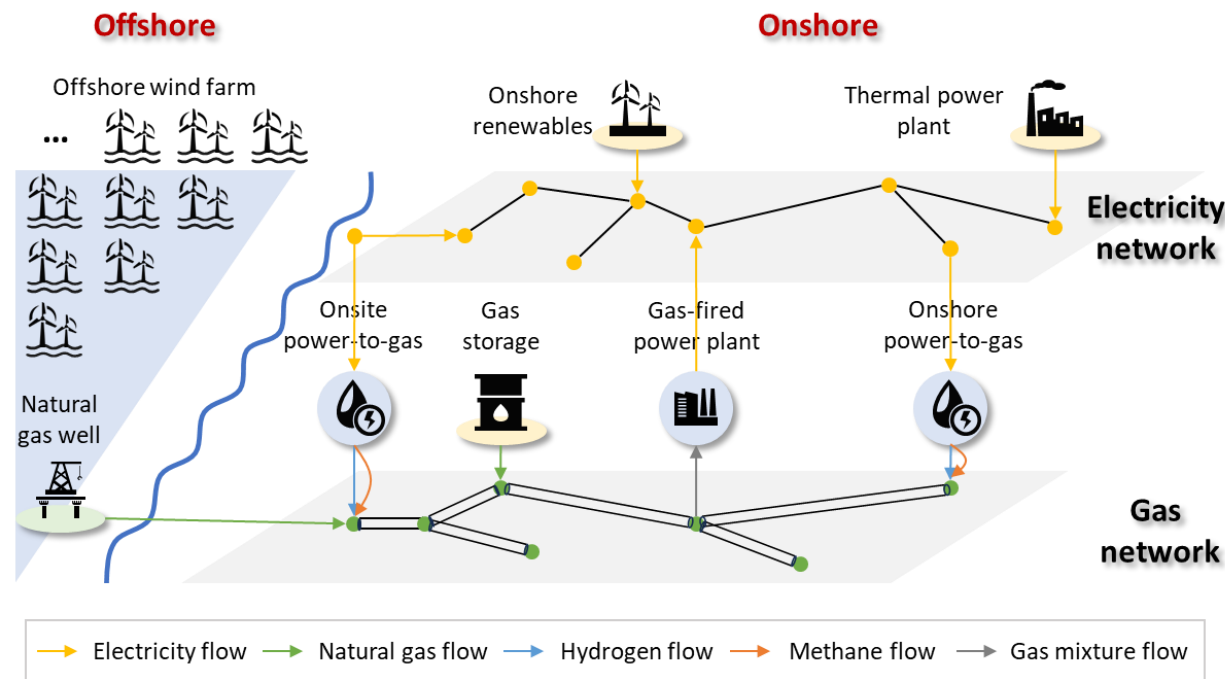
Geo-referenced energy
system structures

Physical, economic, and
carbon emission
parameters of energy
system components

Operation codes of
power and gas systems

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Inputs

- **Objective: minimising total operating cost**
- **Constraints:**



Physical constraints (such
as power line capacity, and
generation capacity)

Flexibility constraints (such
as ramping, on-off status,
and operating reserves)

Security constraints (such
as inertia, must-run-units,
and gas interchangeability)

Hourly integrated energy system optimisation model

Offshore wind
generation, hydrogen
and ammonia
productions

Operating strategies of
power-to-gas facilities,
gas sources, renewable
generations, and other
flexible generating units

Operating conditions of
power and gas
networks

Distribution of
hydrogen molar
fractions

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Outputs